



Retail Demand Forecasting with Machine Learning.

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Table of Contents

Declaration.....	5
Executive summary	6
1.Introduction.....	7
1.1 Background of the Study	7
1.2 Problem statement.....	8
1.3 Research Aims and Objectives	9
1.4 Research Questions	9
1.5 Scope of the Study	9
1.6 Structure of the Dissertation	10
Literature review	11
2.1 Introduction to retail demand forecasting	11
2.2 Importance of Demand Forecasting in Retail Supply Chains.....	11
2.3 Traditional Forecasting Methods.....	12
2.4 Machine Learning in Retail Demand Forecasting	13
2.5 Feature Engineering in Retail Demand Forecasting	14
2.6 Factors Affecting Retail Demand	16
2.7 Research Gap	18
3. Methodology.....	19
3.1 Dataset Description	19
3.2 Data Preprocessing	20
3.3 Feature engineering.....	21
3.4 Model development	22
3.5 Hyperparameter Tuning	23
3.6 Model Evaluation strategy	23
4. Exploratory Data Analysis (EDA).....	24
4.1 Daily Sales Trend.....	24
4.2 Monthly Sales Pattern.....	25
4.3 Weekend vs Weekday	26
4.4 SNAP Impact	27

4.5 Event Analysis.....	28
5.Results and Analysis	29
5.1 Model Performance Comparison.....	29
5.2 Interpretation of Forecasting Results	30
5.3 Feature Importance analysis.....	31
5.4 Actual vs Predicted Analysis	32
5.5 Residual Analysis	33
6. Business Insights and Recommendations	34
6.1 Business Implications	34
7. Conclusions.....	35
7.1 conclusion	35
7.2 Limitations of the research	36
7.3 recommendation for future research	36
References	38
Appendix.....	40
Appendix A: M5 Forecasting Dataset and Data filtering.....	40
Appendix B: Data Transformation and Preprocessing.....	41
Wide → Long format transformation.....	41
Merge with calendar dataset	41
Date conversion	41
Temporal feature creation	42
Appendix C: Exploratory Data Analysis (EDA)	42
Appendix D: Feature Engineering.....	45
Sorting for time-series.....	45
Lag feature	45
Rolling mean feature	45
Appendix E: Model Development.....	45
Feature selection.....	45
Train-test split	46
Random Forest Forecasting Model.....	46
XGBoost Forecasting Model	46
Optimized XGBoost Model	47

Appendix F: Model Evaluation and Results	47
Random Forest Model Evaluation	47
XGBoost Forecasting Model evaluation	48
Actual VS predicted sales.....	49
Prediction error distribution.....	49
Appendix G: GitHub Repository.....	50
Appendix h: AI use declaration	50

Declaration

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Executive summary

Retail demand forecasting is very important for all retail companies as it allows them to make informed decisions about their inventory levels, and reduce the risks associated with stockouts or overstocking. Retailers need accurate forecasts to be able to react quickly to changing demand patterns especially in fast-changing markets like those found in grocery stores where demand varies greatly by day. The objective of this research project is to build a retail demand forecasting model that uses machine learning techniques to forecast future demand for foods at a Walmart store (CA_1) within the M5 forecasting data set, and to test its accuracy with actual demand.

A structured methodology was employed throughout this research project. The first step involved pre-processing the data to transform it into a suitable format for forecasting purposes. An exploratory data analysis (EDA) was then undertaken to understand the trends in sales behaviour, seasonality, fluctuation in demand and shopping patterns of customers. Based upon the insights derived through EDA, a series of new features called "feature engineering" were developed based on historical demand values, temporal attributes and business-related characteristics. These included features known as lag_7, rolling_mean_7, day of week, month, weekend indicator, SNAP program indicator, and event attribute. These attributes were used in order to enhance prediction accuracy.

In addition to developing these new attributes three different machine learning models were also developed and tested. They were random forest, XGBoost and optimized XGBoost. The performance of each model was assessed using mean absolute error (MAE), root mean squared error (RMSE) and coefficient of determination (R^2). Findings showed that while the random forest and XGBoost models produced similar results, XGBoost performed slightly better. Furthermore, when the optimized XGBoost model was applied to the CA_1 Walmart store's FOODS product category an R^2 score of .6863 was obtained along with an MAE of 1.3010 and RMSE of 2.3899. In conclusion, however, the differences among the various models' performances were somewhat modest.

In addition to comparing the relative performance of each model, this research identified some key factors that significantly impacted forecasting performance. As expected, historical demand features such as lag_7 and rolling_mean_7 exhibited the most significant positive correlation with improved forecasting performance. This result supports previous studies indicating that past sales history are critical predictors of future demand. Although business related features such as SNAP Program Indicators and Event Variables provided additional predictive power to forecasting performance they accounted for a much smaller proportion of explanatory power than did historical demand features.

There are some obvious limitations to this research. For example, the findings are specific to only one Walmart store and only one product category. Thus, there is no guarantee that the results will generalize well to other retail contexts. Also, this research only examined forecasting horizons up to one day ahead rather than longer term horizons. Lastly, hyperparameter optimization was done manually because of both time and computational constraints.

Despite these limitations this research demonstrated that machine learning-based forecasting methods can provide reasonable support for short term retail demand forecasting and inventory control in the context of food retail.

1.Introduction

1.1 Background of the Study

The use of Retail Demand Forecasting is very significant in the retail industry since it enables businesses to predict how much of their goods will be demanded in the future so they can better manage their inventory. This kind of demand varies frequently based on the behaviour of customers, seasonal changes, holiday seasons, marketing campaigns and other outside influences. As a result of the unpredictability, many retailers experience challenges such as stock outages, overstocking and waste of products because they were unable to determine how much of the item would be needed. When accurate forecasting occurs, companies can decrease their uncertainty, allow them to develop a better inventory management system and improve the efficiency of their operations (Chopra & Meindl, 2019, pgs. 184-190).

Retailers produce an enormous number of sales records each and every day. With all those sales records generated daily, comes the potential for the application of both predictive analytics and machine learning approaches. Moving Average and ARIMA have been two common forecasting techniques used for decades by retail organizations to predict demand. These types of forecasting techniques however do not work well when demand has complex characteristics or when the demand is non-linear and impacted by multiple interrelated variables. Many researchers agree that machine learning techniques have become increasingly valuable to businesses to predict demand because they are able to discover hidden patterns and relationships, learn from past trends and improve the accuracy of their predictions even though the data set contains thousands of entries and is highly structured (Makridakis et.al., 2022, pgs. 1348 – 1352).

Two popular machine learning algorithms include Random Forest and XGBoost. Both Random Forest and XGBoost have proven themselves to be effective at predicting outcomes in complex data sets and identifying non-linear relationships between different elements. Specifically, XGBoost's sequential learning capabilities enable it to continually improve upon its forecasting results after each prediction error which provides a unique ability to create highly accurate forecasts in structured data sets (Chen & Guestrin, 2016, pgs. 785-794). Therefore, machine learning-based forecasting can provide real-world assistance to retail businesses in terms of enhancing their inventory planning and estimating demand.

This research project aims to use the M5 dataset to conduct retail demand forecasting. The scope of this study will focus on one Walmart location (CA_1), specifically one product category (FOODS) and utilize machine learning techniques to forecast short-term retail demand and compare the performance among models.

1.2 Problem statement

Despite advances in technology, there are still major challenges to providing accurate forecasting of retail demand. Customer purchasing behaviour is constantly changing which makes it difficult to predict with certainty what consumers will want in the next period of time. Additionally, numerous internal and external factors influence consumer purchases in retail markets. Food retail demand uncertainty is particularly high because most food items are sold in small quantities on a frequent basis, perishable and extremely sensitive to rapid behavioural changes. If poor forecasting results occur, then it typically leads to issues related to inventory shortages, excessive storage costs and product waste which directly impacts retailer profitability and operational effectiveness (Chopra & Meindl, 2019, pgs. 184-190).

While traditional forecasting techniques have been utilized extensively throughout history in retail demand forecasting, they often lack sufficient ability to recognize non-linear demand behaviours and interact amongst various components. Prior studies generally evaluated total forecasting performance with no regard to analysing business-related features like government programs and special events occurring within a specific retail environment. There exists a clear necessity to assess whether utilizing combinations of historical, temporal and business-related attributes through machine learning methodologies could positively impact the performance of short-term retail demand forecasting.

1.3 Research Aims and Objectives

Overall objective of this research project was to produce a retail demand forecasting model based on Machine Learning that would allow the prediction of short term food demand within the Retail Sector.

In order to accomplish the overall goal of developing a retail demand forecasting model, several specific research objectives have been established:

- To analyze and preprocess the retail sales data for the M5 Forecasting Dataset.
- Conduct an Exploratory Data Analysis (EDA) on the retail sales data to identify seasonal behaviour, time patterns and the characteristics of demand.
- Develop and compare two different types of machine learning models; namely Random Forest and XGBoost for their ability to accurately forecast retail demand.
- Utilize feature engineering techniques including historical, temporal and business-based variables in order to extract relevant information from the data that will contribute to improving the accuracy of the forecasts produced by the machine learning models.
- Evaluate the performance of each machine learning model using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and Coefficient of Determination (R^2).
- Determine if the inclusion of business-related features improves the accuracy of short term retail demand forecasting.

1.4 Research Questions

This research seeks to answer three research questions:

- How effective are Machine Learning Models at producing accurate predictions for short-term retail food demand?
- When comparing Random Forest and XGBoost models which type of machine learning model produces more accurate forecasts for retail demand?
- What impact does the inclusion of business-related features (i.e., SNAP Program participation and events) have on the accuracy of short-term retail demand forecasting?

1.5 Scope of the Study

This research is focused exclusively on the M5 Forecasting Dataset and therefore, all data used for the purpose of this research has been extracted from this single source. This research is further restricted to the retail data collected from one store location (CA_1) and one retail category (FOODS). Additionally, this research only explores the application of Machine Learning models for Short-Term Retail Demand Forecasting and

therefore, only utilizes a 1-Day Ahead Forecasting Horizon. Only Random Forest and XGBoost Machine Learning Models have been applied to the data in order to compare their respective performances. While this research offers valuable insights into food retail forecasting it should be noted that due to these restrictions, the results may not be applicable to other locations, product categories or Long Term Forecasting Scenarios.

1.6 Structure of the Dissertation

The seven chapters that make up this dissertation are as follows:

Chapter 1 is an introduction to the study background of this dissertation; problem statement; objectives of the study; questions of the research and scope of the study.

Chapter 2 is a review of the literature on all previous studies concerning Retail Demand Forecasting; traditional methods of forecasting; machine learning techniques for forecasting; feature engineering; factors influencing retail demand; and gap in research.

Chapter 3 details the methodology used in the entire dissertation including dataset description; data preprocessing; feature engineering; model development; hyperparameter tuning; and evaluation strategy for models developed in this dissertation.

Chapter 4 outlines the exploratory data analysis performed during this research to determine characteristics of sales behavior, seasonal patterns, demand variability and the interrelationships among the variable influences on modeling.

Chapter 5 includes results and analysis from this dissertation including comparison of performance among models; interpretation of forecast results; importance analysis of features; actual versus predicted values comparison; and residual analysis.

Chapter 6 offers insights into business applications of and recommendations based upon the findings of this research and describes the practical implications of applying machine learning-based forecasting in retail demand planning processes.

Chapter 7 is the conclusion of this dissertation, which summarizes the major findings, identifies limitations to the research presented herein, and offers suggestions for future research.

Literature review

2.1 Introduction to retail demand forecasting

The term retail demand forecasting means predicting the customer demand, sales quantity and requirement of the inventory of the future. By using historical sales data, patterns, trends the retailers can forecast the demand which helps them in the preparation of inventory, reduce uncertainty and improve operational planning. According to (Makridakis et al. 2022) forecasting helps the organisations to make proper decisions under uncertain conditions by using historical information.

Retail demand changes because of customer preferences, seasonality patterns, events and other external features without forecasting business may face stockouts, overstocking and wastage problems especially in food category. Previous studies suggest that forecasting is important for inventory and supply chain management (Chopra and Meindl, 2019, pp. 184–190).

The traditional forecasting methods are widely used to forecast the demand. Since the retail data is large and complex nowadays retailers have started using data analytics and machine learning to improve the accuracy as the studies have shown machine learning can learn complex relationships from which are useful for predictive modelling (James et al., 2021, pp. 327–350).

This research explores retail demand forecasting by using machine learning techniques to evaluate forecasting performance in retail sales prediction.

2.2 Importance of Demand Forecasting in Retail Supply Chains

The importance of forecasting the demand in the retail supply chain is huge as it helps the businesses to estimate the future demand for the products, manage inventory, allocate resources, improve overall operational planning. Purchasing, stocking and distribution process in retail sectors heavily depend on the accurate forecasting of demand. According to (Chopra and Meindl 2019, pp. 184–190) forecasting improves the efficiency of supply chain, inventory planning and operational performance.

The research indicates poor forecasting led to overstocking meaning excess inventory which can increase storage costs and wastage, Stockouts leads to lost sales and lower customer satisfaction. Previous studies suggest that this is very important in food retail as the products are perishable and sensitive to demand fluctuations (Chopra and Meindl, 2019, pp. 184–190).

Several authors argue that the forecasting the demand actually supports better decision-making, inventory control and demand planning. The firms are benefited by accurate forecasting to reduce uncertainty and improve business efficiency. Retailers

have started to heavily rely on data-driven forecasting approach to stay competitive (Makridakis et al. ,2022).

The forecasting has become significant in the modern days as the researchers have developed various forecasting techniques to support various types of data including traditional statistical methods and machines learning approaches. This helps the retail firms to predict the demand to an extent which help them to make informed operational decisions to improve the efficiency.

2.3 Traditional Forecasting Methods

Before the machine learning methods were popular , the retail demand forecasting was heavily dependent on the traditional methods which involves the usage of historical sales observations. This helped the retailers to identify the patterns and predict demand from the available data. These methods were popular because it was simple to understand , easy to implement. The firms preferred these methods because the forecasting output was structured and easily interpretable which helps them to have efficient inventory management and operational planning (Hyndman and Athanasopoulos ,2021, pp. 1–25).

The most commonly used traditional forecasting method is Moving average method. The model works in a basis of calculating the average of previous observations over a given period and predict the demand of the future period. It smooths fluctuations by not relying on one previous sales value. This helps the firms to identify the general trend of the demand as it reduces the short-term fluctuations in demand. It only performs well when the demand is relatively stable and not a lot of fluctuations over time. It tends to react slow to sudden changes in demand , seasonal spikes and unexpected behavioural changes in the customers(Hyndman and Athanasopoulos,2021, pp. 177-185).

Another commonly used forecasting approach is exponential smoothing. This method works in a comparatively different way to moving average method as it prioritizes the recent observations compare to the older observations. The reason behind this approach is that the customer purchasing pattern on a recent time may better represent the near future demand patterns. Several authors argue that this method improves the forecasting as it generally adapts more quickly to the changes in demand happened in the recent times. Though this method is useful when demand changes over gradually it still struggles when the demand becomes too complicated, non-linear or influenced by the multiple external factors(Hyndman and Athanasopoulos ,2021, pp. 186–220)

ARIMA model is the most widely used statistical forecasting method. It is mainly used for time-series forecasting as it finds the relationships between historical observations across a period of time. Studies shows that ARIMA performs well when the

demand pattern remains stable and repetitive over time as it tends to capture trend , seasonality patterns and autocorrelation. Since the demand for the retail sectors is often affected by various different factors like customer behaviour , promotions , holidays and external market conditions ARIMA may struggle as these factors fluctuate the demand pattern a lot and make them highly non-linear. (Hyndman and Athanasopoulos, 2021, pp. 241–285)

Though the traditional forecasting methods are widely used in the retail sector to predict the demand pattern they mainly depend on stable and repetitive historical behaviours. The retail demand in the modern times had become more complex , large and dynamic where several factors affect the purchasing patterns of the customer. Researchers indicates that these factors could struggle to capture non-linear and relationships and multiple factors which interacts with each other affecting the retail demand. Due to these limitations researchers started exploring machine learning techniques to improve forecasting accuracy as they tend to identify hidden relationships and complex patterns (Makridakis et al., 2022)

2.4 Machine Learning in Retail Demand Forecasting

As mentioned before the traditional forecasting methods perform well if the demand pattern is stable, but the modern retail demand became more complex dynamic and nonlinear for traditional methods to stay relevant. As factors like customer purchasing behaviour , seasonality , holidays, promotions interact with each other makes the forecasting process a bit difficult for these methods to handle. Previous conducted research indicates that machine learning methods became increasingly useful and popular because they can identify those hidden relationships and non-linear patterns from a large dataset (Makridakis et al. ,2022). Machine learning models perform better than the traditional ones as it learns directly from the historical data and improve predictive performance by identifying patterns automatically. Several authors argue that machine learning performs really well particularly in retail demand forecasting where the demand fluctuates frequently (James et al. ,2021, pp. 327–350).

Unlike traditional statistical forecasting , machine learning does not just assume of stable linear relationships it looks out for non-linear relationships, find hidden pattern , process high -dimensional data to forecasting the demand. As studies suggest that machine leaning methods often outperform traditional methods when the dataset is large and has complex behavioural patterns. It is also more flexible than the traditional methods when forecasting retail demand due to its fluctuations (James et al., 2021, pp. 327–350; Makridakis et al., 2018). Machine learning models can include multiple explanatory variables at a same time by increasing it predictive capability (James et al.,2021, pp. 327–350).

Random forest is one of the widely used machine learning models for predictive modelling. It is an ensemble (grouping) learning method which combines decisions of multiple trees to improve the accuracy of its prediction (Breiman ,2001, pp. 5–32). It reduces the overfitting risk by combining multiple trees rather than relying on just one tree. Studies have shown that Random Forest performs well if the data has non-linear relationships , high in volume, multiple interacting variables. The major advantage in using Random Forest model is that it can also provide feature importance analysis which helps in understanding which influence prediction the most. Several studies suggest that Random Forest model is useful in forecasting retail demand due to its robustness (ability to maintain high performance and accuracy) and its stable predicting behaviour (Breiman, 2001, pp. 5–32).

The another widely used machine learning model is XGBoost for forecasting and predictive analysis. Unlike ensemble learning like Random Forest model , XGBoost follows sequential learning method. In this method the model builds new trees to correct the previous predictive errors thus it often improves the predictive performance (Chen and Guestrin (2016, pp. 785–794)). Studies have shown that if the data is too structured XGBoost will most likely outperform traditional forecasting methods and many tree-based models. It became popular among the research because it handles non-linear relationships , missing values and if the data has complex interaction in variables. Research indicates that XGBoost performs well than other tree-based models in forecasting competitions and predictive modelling (Makridakis et al. ,2022, pp. 1348–1352).

Although machine learning methods have a lot of advantages in improving the forecasting performance it also has limitations like it require significant amount of time, memory due to its complexity , can be harder to interpret and heavily dependence on data to forecast. It also depends on the quality of the data , feature engineering and hyperparameter tuning for its efficient performance. (James et al. ,2021, pp. 333–350).

Previous literature suggests that creating meaningful predictive variables through feature engineering improves the performance of the machine learning models as it heavily depends on the input variables for its forecasting efficiency.

2.5 Feature Engineering in Retail Demand Forecasting

Previous literature suggests that conducting feature engineering is an important process in predictive modelling and forecasting. Feature engineering means transforming raw data from the dataset into meaningful variable which helps in the process of forecasting the demand. Several authors argue that the machine learning models heavily depend upon the feature engineering to improve its forecasting accuracy rather than just relying on complexity of the model as variable derived from a

good feature engineering helps models in identifying hidden relationships , reduce unwanted information . Research indicated that a feature engineering could result in poor forecasting performance even when advance machine learning models are used (Kuhn and Johnson ,2019, pp. 3–25).

One of the most common methods of developing features through feature engineering for retail demand forecasting is through the utilization of lagged values. Historical sales data from previous periods are used to predict future demand because studies have shown that past demand behaviour often influences future purchasing patterns. In an example of weekly demand repetition , seven-day lag values are used to determine the pattern of demand.

Studies have found that lagged values were extremely helpful in retail forecasting because the purchasing history of customers tends to repeat itself throughout the course of time . Additionally, lagged values allow machine learning algorithms to recognize temporal dependencies in demand. As stated by Box, Jenkins & Reinsel (2015, pp. 10-30), this method helps create lag relationships and demonstrates time series dependency. Furthermore, Makridakis et al. (2022, pp. 6-10) demonstrate that using historical observation enables forecasting.

Another widely used method of creating features is with rolling statistics , specifically rolling means. The rolling mean creates an average of demand based on a certain amount of time in the past. Studies have demonstrated that rolling means help smoothen out short term variations in demand, remove noise from data, and help determine short term trends in demand. Multiple authors have indicated that rolling features are superior to individual historical observations when determining recent trends in demand. Because of the frequent changes in demand seen in retail forecasting, rolling features become very valuable. As stated by Kuhn & Johnson (2019, pp. 45-65), these types of features assist in predictive feature generation and are effective when utilizing rolling variables.

Previous literature has demonstrated that both calendar and temporal features can also contribute to retail demand forecasting. Features such as day of week , month , year and weekend will allow models to capture seasonal demand, weekly buying habits and other temporal related patterns. Studies have shown that there is a difference in buying habits between days of the week. By incorporating temporal variables into machine learning algorithms, you can better understand seasonal and repetitive buying patterns. This concept was demonstrated by Makridakis et al. (2022, pp. 1348-1352) by providing examples of variables used within forecasting competitions, as well as temporal forecasting.

Additionally, research indicates that retail demand could also be influenced by external business-related variables. Examples include events , promotions , holidays and government programs. There have been several studies demonstrating that if you incorporate external variables into your model, it will lead to improved forecasting results because there are many factors that affect purchasing behaviour besides historical sales. Therefore, it is essential for organizations to properly engineer their features to increase the accuracy of their forecasts regarding retail demand.

2.6 Factors Affecting Retail Demand

The retail demand is heavily influenced by various factors. Therefore, it is challenging to estimate retail demand precisely. This is due to fact that customer behaviour constantly changes due to an array of internal and external influencing factors. As stated in prior literature, when we do not take account of these factors it could result in lower levels of forecasting precision as well as inefficient inventory planning. Research has demonstrated that demand volatility in the retail sector could arise from the following Seasonality , Holidays ,Customer Behaviour ,External Influences on the Market and Promotion (Makridakis et al., 2018).

Seasonality is one of the main factors that affect retail demand. Past studies have indicated that there are differences in consumer purchasing behaviours at various times of the year as well as their purchasing habits. Temporal patterns greatly affect consumer purchasing behaviour, especially in food retail. In particular, since many consumers regularly buy products and have changing demands for those products; temporal patterns are essential to estimating demand in food retail. Also, weekly recurring patterns and monthly variations in retail sales data make incorporating seasonal and temporal variables into forecasting models necessary. For this reason, some researchers use variables such as day of week, month or weekend indicators to provide a deeper insight into recurring consumer purchasing behaviours as well as seasonal trends in demand. Additionally similar forecasting approaches were applied in forecasting competitions where temporal variables improved forecasting results (Makridakis et al., 2022, pp. 1348-1352).

Holiday and special events are another factor that influence retail demand. Prior research indicates that consumers behave differently than normal on public holidays, festivals, sporting events, and other celebrations. On these days consumption increases and people gather with family and friends. Due to the fact that consumers purchase more groceries and household items for celebrations and family activities demand tends to increase during holiday periods. Studies demonstrate that occasional spikes in sales occur on occasions such as Easter, Super Bowl, and Labor Day which temporarily increase demand for certain products resulting in abnormal sales spikes in

retail stores. Therefore, researchers have included event-based variables into their forecasting models to capture temporary shifts in consumer buying behaviour caused by special events (Makridakis et al., 2022, pp. 1348–1352).

Promotions and discounts are also a Major Factor Which Influence Retail Demand. Prior research indicates that consumers react positively to discount offers and marketing campaigns because price reductions often stimulate greater buying behaviour. Temporary boosts in sales volumes created by promotional activity can cause short term spikes in demand which are difficult to anticipate unless appropriate forecasting variables are used. Several researchers argues that promotional activity has a significant effect on the purchasing behaviour and demand trend of consumers in retail markets. Because of the strong impact promotional activities have on consumer purchasing decisions and sales quantities; promotional activities are typically included as an important variable in forecasts.

Customer purchasing behaviour is also a key driver of retail demand. Prior research shows that customer preferences and shopping behaviour may change due to lifestyle changes, changes in their income levels, increased availability of convenient shopping options and evolving household needs. Authors agree that forecasting purchasing behaviour is problematic because consumer demand can shift rapidly at any time. These unpredictable behavioural shifts contribute to increased forecasting uncertainty and create difficulties in anticipating demand accurately. Therefore, high variability exists within retail demand due to continuous evolution of consumer preferences making it difficult to forecast demand within the retail industry (Makridakis et al., 2018).

In addition, research indicates that government assistance programs can influence retail demand specifically within categories related to food. Increases in purchasing power among low-income households during limited-time government assistance programs can increase grocery purchases. Studies demonstrate that households experiencing changes in purchasing behaviour exist after receiving government support. Therefore, government related programs can serve as additional external forecasting variables specifically for grocery and food demand forecasting purposes (Nord & Prell, 2011, pp. 1-12).

Generally speaking, prior research suggests that retail demand is driven by both historical demand behaviour as well as external factors including seasonality, promotions, holidays, customer behaviour and government support. Given the complexities involved in the interactions of these variables; estimating retail demand is becoming increasingly complex. Due to these complexities, researchers have increasingly started using advanced forecasting methods to improve forecasting accuracy.

2.7 Research Gap

Retail demand forecasting through traditional statistical forecasting methods and machine learning approaches has received significant attention in prior literature; however, some limitations continue to affect research in this area. The majority of prior studies have centred around either increasing forecasting accuracy or comparing various predictive techniques for estimating demand. Machine learning models, specifically Random Forest and XGBoost models, have provided superior forecast performance than many other types of models, including those based on time series methodologies, when large amounts of structured data with complex inter-variable relationships exist (Chen & Guestrin, 2016, pp. 785-794). Similarly, forecasting competitions and prior studies have demonstrated that advanced forecasting methods can improve demand prediction performance in retail environments (Makridakis et al., 2022, pp. 1348–1352).

Despite these contributions to the field of retail demand forecasting, several gaps remain in the prior literature. The majority of prior studies have primarily focused on overall forecasting performance rather than investigating a specific retail category or conducting a store level analysis. Retail demand can be quite variable between stores and product categories and therefore generalizing demand behaviour across all possible retail settings using single generalized forecasting method may not adequately capture the complexity of demand behaviours at the retail level. Additionally, prior literature has very little discussion about how business-related variables contribute to demand forecasting in retail. Variables such as special events, government support programs and external influences on consumer purchasing behaviours were not considered as part of the model used to develop demand forecasts. While machine learning-based forecasting models have been well-discussed within the body of literature, there is a lack of knowledge surrounding how the inclusion of business-related variables improves demand forecasting performance in the context of forecasting food retail demand .

Therefore, this research aims to address these gaps by conducting a more focused retail demand analysis at a single Walmart store (CA_1), using one product category (FOODS) from the M5 Forecasting Data Set. A comparative evaluation will be conducted to compare the forecasting performance among three types of machine learning-based forecasting models: Random Forest Models, XGBoost Models and Optimized XGBoost Models. Historical demand variables and temporal features will be included as input variables into each model type. Business-related variables such as SNAP program indicator and event variables will also be incorporated into the models. Finally, this research evaluates short term retail demand forecasting over a 1-day-ahead forecasting horizon to aid in supporting inventory planning and operational decision-making in food retail.

3. Methodology

In this chapter I will tend to explain the methodological process which used to create this research. I will go through what dataset I selected for this research , the methods for preprocessing , EDA to understand the basic theme of the data which leads to choose and develop models in the further stages. Methodology is an important process as it ensures transparency and reproducibility of research as said by James et al. that predictive modelling requires a structured workflow involving those previous steps which I had mentioned above. (James et al., 2021, pp. 1–5)

Quantitative research approach is followed to conduct this forecasting study by using numerical and historical sales data , machine learning models such as Random Forest and XGBoost are used to discover pattern. As M5 competitions dataset is used this research adopts similar predictive forecasting objective as found in (Makridakis et al., 2022). The roadmap of the research starts dataset loading , cleaning and setting up the data (preprocessing) , conduct explanatory data analysis (EDA), feature engineering , model development and evaluation of the models and its results.

3.1 Dataset Description

This research uses M5 forecasting dataset which is originally developed for M% forecasting accuracy competition and its publicly available through Kaggle. It contains Walmart (a huge retail brand in America and Canada) retail sales data , historical daily sales , multiple product categories which available on multiple stores and different states. The reason why I choose this dataset because its real-world dataset and its huge which is pretty suitable for the machine learning forecast. It also helps me capture realistic retail behaviour. (Makridakis et al., 2022, pp. 1348–1352)

The sales data were the primary forecasting target for this research as the dataset contains daily sales across time periods , demand based on items and daily sales observations too. The secondary variable was time (calendar) because it easily reveals pattern in a data. The data had a separate CSV file named calendar which actually contains weekdays , special events , SNAP programme indicators (Makridakis et al., 2022, pp. 1348–1352). With these data we created dayofweek , month , year, weekend, snap CA , event names. The purpose behind this is to organise the data for further analysis on sales which provides temporal context and external demand influences.

Since it's a huge data containing literally every product sales on a daily basis across the retails stores which are in different states , The data is restricted to one particular store -CA 1 and one particular category- FOODS (as it purchased in frequent

basis and will be highly relevant for inventory planning) to make the analysis feasible , reduce computational burden and simplify models.

This restriction allowed the data to have deeper analysis , feature engineering and performed well in the models with detailed interpretation without any struggles. I tend to choose focused analysis over analysing the whole dataset because it creates accurate results which is crucial for inventory management (Chopra and Meindl, 2019, pp. 184–188).

The data contains approximately five years of daily observations across the retail stores starting from 2011 to 2016. This long period of data actually helps to find seasonality patterns , fluctuation in the demand , recurring purchase pattern so on. It also helps in models because predictive models perform better when sufficient historical observations exist (James et al., 2021, pp. 15–20).

3.2 Data Preprocessing

Unprocessed data cannot be used for analysis because it has missing values , bad structure, unnecessary complexity to the data which leads to the inaccurate results. That's where preprocessing comes in it improves the data quality , prepare the data for modelling and helps avoid inaccurate predictions.

As said earlier the original dataset had been filtered to CA 1 store in particular food category which is done to remove all the unnecessary observations and making forecasting actually possible (Chopra and Meindl, 2019, pp. 184–188).

The original dataset was in a wide format storing each day as a separate column representing daily sales for a product. This structure was so complicated and difficult for machine learning , time-series forecasting and temporal analysis as the model need date and sales combined not thousands of day columns. I transformed this wide format to long format using reshaping process , instead of having many daily columns I converted them into single date column and single sales column. By doing so , now each row represents one product , one day and sales value on that particular day which became like a product -day observation.

This transformation was necessary because it enabled the time-series analysis , date-based forecasting and machine learning implementation without restructuring lag variables impossible , rolling averages impossible and forecasting difficulty. After converting to long format now the model can easily understand the relationship with date and sales , it also made possible to create the temporal features such as day of the week , month ,year ,weekend and so on which helps in identifying the patterns in the data. It also enabled the lag variable which allows the model to see the past sales behaviour. In our case if sales were high last week the model can use this information to for the future prediction. Further it enables rolling mean 7 which smooths short-term

fluctuations helps to understand the demand trend better , identify average weekly behaviour. Restructuring helped transform raw data into suitable format which helps machine learning forecasting.

3.3 Feature engineering

The raw sale data is alone is not sufficient to start modelling , The retail demand is affected by the time patterns , historical behaviour and external influences and thy is why we conduct feature engineering to improve the model performance as a whole , to extract meaningful patterns and to transform all the raw sales data into predictive variable to enhance the performance of the models. The feature engineering actually helps the model by noticing the hidden relationship and improve the forecasting accuracy (James et al., 2021, pp. 297–305).

I first created temporal features , The day of week was created to get the weekly purchasing behaviour it because the EDA showed weekly recurring demand patterns, and this allows the model to identify the weekday differences. Month feature was created to capture the seasonality is also discussed in the EDA chapter which is to get the seasonality trends into account as it showed moderate seasonal variation. Then Year feature was created to capture the long -term temporal change , the reason behind this is to find whether the demand evolves over time or not. Weekend feature was also created due to significant increase in the sales during weekends and the purpose is to get the behavioural demand change. These temporal features were directly influenced by the findings of EDA.

Secondly, I created Historical Demand Features, Lag 7 was created to get the sales from exactly 7 days before to identify the weekly demand repetition. As EDA showed weekly recurring spikes the past sales often influence the future demand. Rollin mean 7 was created to calculate the average sales over a seven day period . It helps smooth out the fluctuation in the data , reduce noise and identify if there is any short-term trend. Later to avoid leakage I shifted the rolling statistics to go one period backward before calculation to use only historically available information for prediction. This make sure that the future information was not introduced unintentionally into the process of forecasting. Historical variables improve predictive modelling and temporal forecasting (James et al., 2021, pp. 297–305).

Finally , the business features such as SNAP CA and event encoded was created as EDA showed higher demand during SNAP periods and increase in sales during special events. These are created to capture changes in demand due to external policy and event driven purchasing behaviour. As the event is categorical label encoding is used to convert text categories into numerical values. Though label encoding may create an order between categories , the tree-based models are usually less affected by this problem which are used in this research. (Prokhorenkova et al., 2018, pp. 6638–

6648). These proper engineered features convert the raw sales data into meaningful predictive variables to implement the machine learning models.

3.4 Model development

The reason behind to go with machine learning models is purely because the traditional forecasting methods may struggle with the complex retail demand data , analysing non-linear relationships and changing demand behaviour on variables. Whereas the machine learning models can identify hidden relationships , learn and update from historical patterns , improve predictive accuracy. In this research , I used Random Forest and XGBoost to compare the predictive performance as tree-based models are suitable for prediction tasks , if the dataset is too complex and contains non-linear problems (James et al., 2021, pp. 327–350).

The Random Forest is selected as my baseline forecasting model. It is a ensemble learning model which combines multiple decision trees to improve prediction stability and also reduces overfitting risk. The model works by combining predictions from multiple trees and averaging the output. The purpose of this in the research is to establish initial forecasting benchmark and compare it to model which are created later. It is chosen to handle the nonlinear relationships , pretty suitable for retail forecasting and also helps in interpreting feature importance(Breiman, 2001, pp. 5–32).

After Random Forest , XGBoost model was implemented. This is a gradient boosting model , often provides higher predictive accuracy and it builds model in an step by step fashion (sequential learning) it learns from its previous mistakes. This characteristic often enables XGBoost to outperform traditional tree-based approaches in forecasting tasks. It is mainly useful because it can handle complex interactions , also captures non-linear patterns and performs well on structured datasets (Chen and Guestrin, 2016, pp. 785–794).

At last, I enhanced XGBoost with the business features to see if there is any improvement from the previous model as it is created using temporal features and historical demand features. The business-related variables which were added to this model are SNAP CA and event encoded. It is created to address the possibility that events , government programmes , consumer behaviour can also influence retail demand. (Chopra and Meindl, 2019, pp. 184–190). After the development of models , hyperparameter tuning and performance evaluation was conducted to increase the forecasting performance and compare model effectiveness.

3.5 Hyperparameter Tuning

The default model parameters may not provide the best performance, so hyperparameter tuning is done to improve forecasting accuracy. The purpose of this was to further reduce the prediction error, improve the model learning and avoid overfitting as the model performance depends upon the parameter selection (James et al., 2021, pp. 333–340).

The XGBoost model parameters are adjusted through manual experimentation.

- `n_estimators` (number of trees) = 200
- `max_depth` (complexity of the model) = 8
- `learning_rate` (speed of learning) = 0.05
- `subsample` = 0.8
- `colsample_bytree` (reduce overfitting) = 0.8
- `min_child_weight` (improve model robustness) = 3

The performance of this was evaluated using RMSE, MAE, R^2 as they provide comprehensive evaluation of forecasting performance. Due to computational and time limitations, an exhaustive search approach such as grid search was not implemented. I used the forecasting metrics as mentioned above to evaluate the final model performance and to compare with the previous model.

3.6 Model Evaluation strategy

After developing the model it is necessary to evaluate it to assess the forecasting accuracy, compare it with the different models to identify the best performing model of the research. I used performance metrics to compare random forest and XGBoost models output (James et al., 2021, pp. 254–260).

As said above the used metrics are

- Mean Absolute Error (MAE) which helps to understand prediction accuracy by measuring forecasting error on an average (lower MAE = better model).
- RMSE (Root Mean Square Error) the purpose of this is to identify how reliable the forecasting is. This metric penalises larger errors more strongly. (Lower RMSE better.)
- R^2 this measures how much variations explained. Which helps to understand the explanatory power. (Higher the R^2 the better) (James et al., 2021, pp. 254–260)

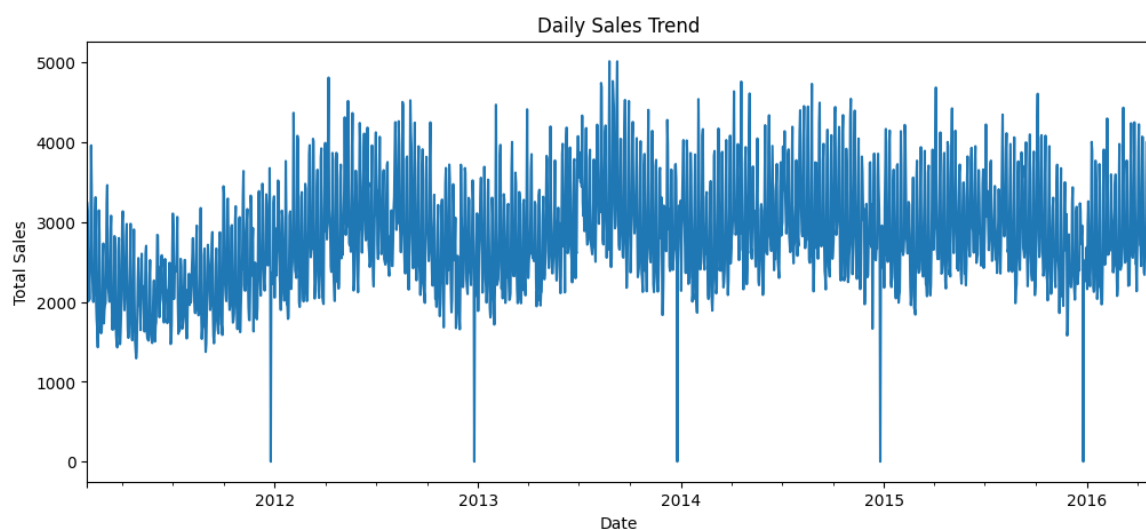
I also did the Visual evaluation to convey the model performance visually. Actual vs Predicted sales graph was done to compare the prediction behaviour. Residual

distribution is analysed to identify forecasting error patterns and the residual was concentrated around zero which indicate better forecasting performance of the model. This helped me to compare different models and identify the most effective approach for retail demand prediction.

4. Exploratory Data Analysis (EDA)

In research the EDA is conducted to understand the data behavioural patterns , identifying sales patterns , seasonality observations , understand when demand fluctuates, exploring and utilizing the relationships between variables. This process will hugely help to understand the data before the model development and make better feature engineering decisions. It is conducted not only to understand the descriptive narrating but also to motivate the modelling decisions later in the process (James et al., 2021, pp. 1–5).

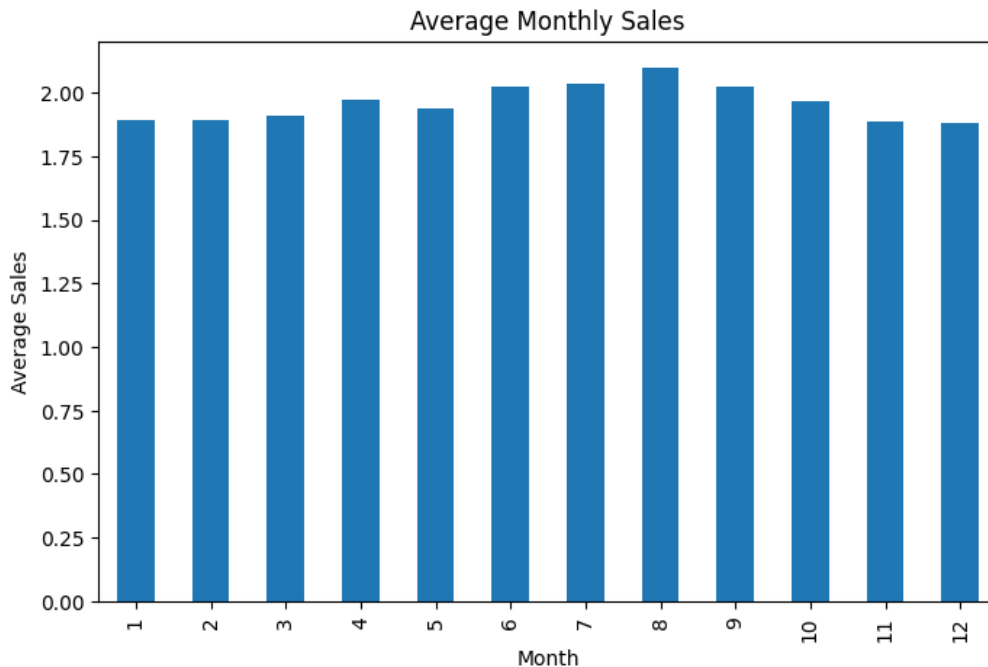
4.1 Daily Sales Trend



For the first graph I did a daily sales trend visual to understand the fluctuations in sales , recurring patterns , overall movement in demand on a daily basis. The graph shows specifically that sales fluctuate heavily across the whole period and noticeable spikes recurring on a weekly basis is clearly visible too. The demand usually ranges between 1500- 4500 in volume. The other thing to be noted is the there is a slight increase in the daily sales between 2012 -2014 and also before stabilising you can find several sharp drops to almost zero these may be due to holiday closures , missing demand periods and unusual operational events. These suggest strong temporal dependency ; there is a recurring behaviour in demand. These things directly motivate

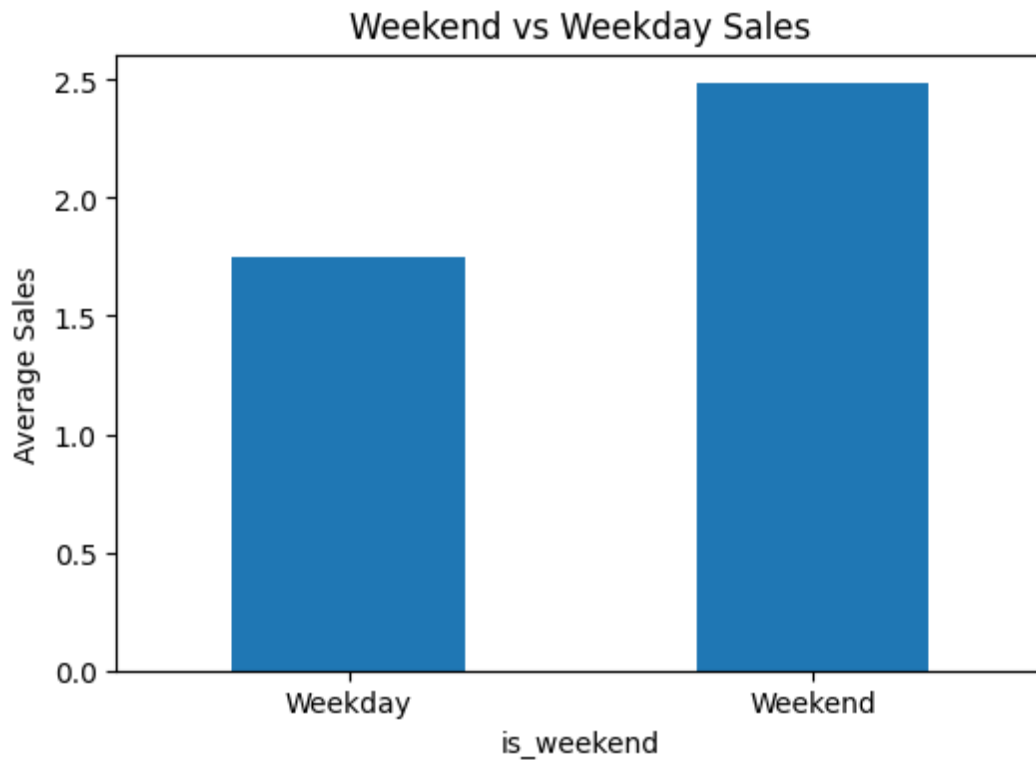
lag 7 and rolling mean 7 because the past sales behaviour will be useful in predicting future demand. It is seen that the weekly repeating pattern justified lag 7.

4.2 Monthly Sales Pattern



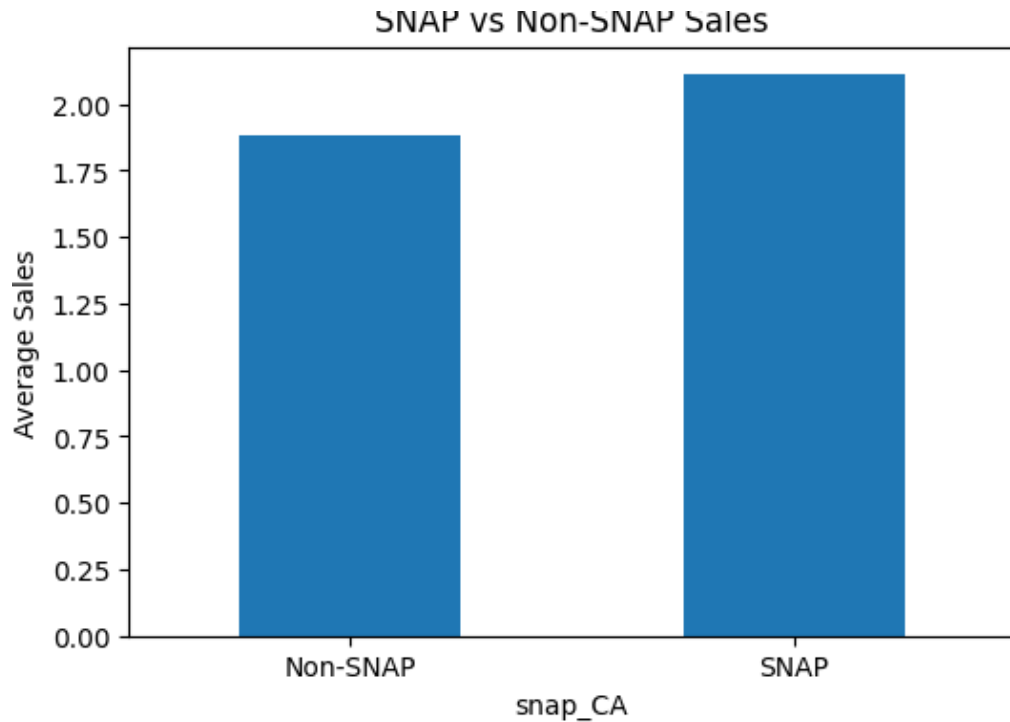
Secondly , the monthly sales pattern is analysed to identify the pattern in seasonality demand. Though the seasonal variation exists way moderately , the monthly demand remains relatively stable. The highest average sales were found in the month of august around 2.1 average sales and the lowest is observed during January , February and December showing approximately between 1.88-1.90. This analysis denotes that there is mild seasonality, and the purchasing behaviour of the customer varies during the year which shows demand changes across months. This helped the modelling phase to have month feature which allows the model to capture monthly seasonal variation.

4.3 Weekend vs Weekday



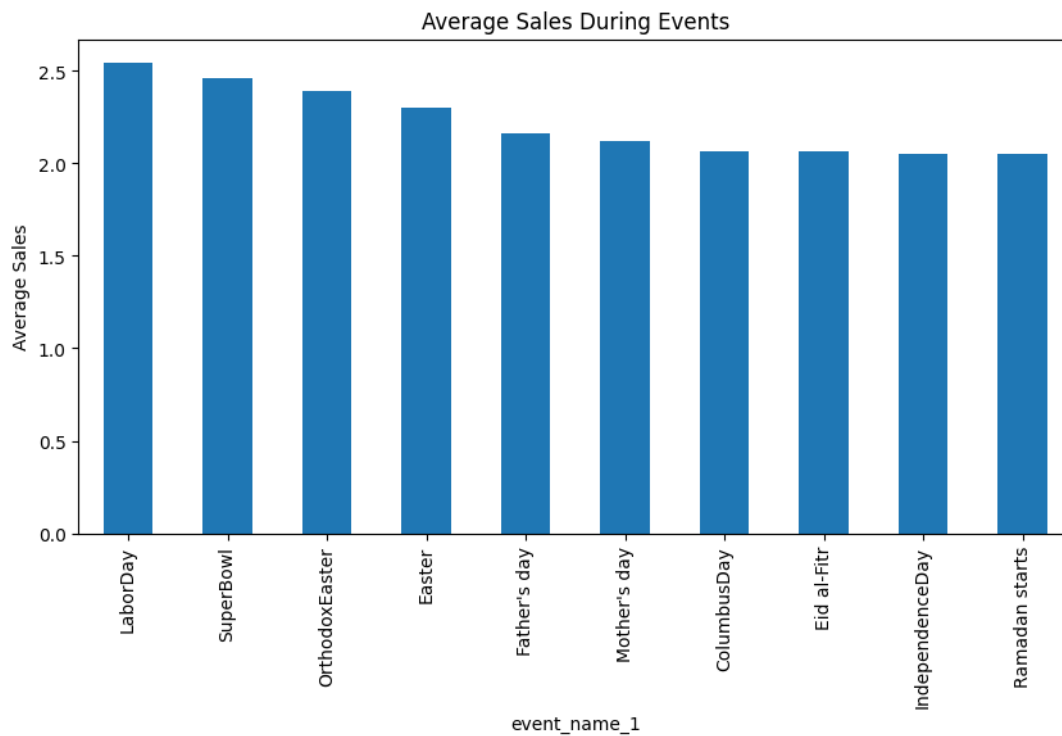
On the third graph , we analysed the demand between weekdays and weekends which also helps to understand another perspective of customers on their purchasing behaviour. The result was so obvious which clearly stated the weekend sales(approx. 2.5) are significantly higher than the weekday sales(approx. 1.75).The possible reason for this difference can be that the customers have more shopping time and higher household activity are done during the weekend days. This hugely impacted the modelling because the weekend effect is clearly visible.

4.4 SNAP Impact



For the fourth graph snap programme analysis was done , which analyses the during SNAP and non-SNAP periods. It helps us to understand the influence of government assistance programme. It shows that the SNAP periods (approx.2.12) show higher sales than the non-SNAP periods(approx. 1.88) which in general term means that the government food assistance program may influence the food category (groceries) purchasing behaviour in customers. It influences the modelling to have a snap ca feature because the business-related external factors may influence demand forecasting.

4.5 Event Analysis



For the last graph we did the event analysis which find any pattern or changes on the event days such as labour day , super bowl so on . In this analysis it is found that the highest average sales were during labour day followed by Super Bowl , Orthodox Easter , Easter. On the other hand, the lowest average sales were observed on Independence day and Ramadan starts. However, there is an interesting find that all event periods shoes elevated demand above 2.0 average sales. This means that the specials occasions influence purchasing behaviour especially on holidays , celebrations and gatherings. These finding influenced modelling to include event names and event encoded to capture event-driven demand variation.

The EDA revealed that there is strong demand fluctuations on a daily basis , weekly recurring sales behaviour, mild seasonality , stronger weekend demand , the influence of the SNAP programme is real and there is a noticeable increase in demand on the event days. These justify in implementing “lag_7” , “rolling_mean_7” , “ month” , “is_weekend” , “snap_CA” , “event_encoded”.

5.Results and Analysis

5.1 Model Performance Comparison

The purpose of the model comparison is to compare different models analyse each model's performance and identify the best performing model while assessing the impact of business features. Three models have been evaluated (Random Forest , XGBoost , Optimized XGBoost) using metrics such asMAE, RMSE and R square. As mentioned earlier lower MAE/RMSE means better performance of the model and High R square mean better performance too.

Random Forest achieved

- MAE = 1.3079
- RMSE = 2.4034
- $R^2 = 0.6827$

It shows a reasonable forecasting performance, and the R square indicates approx. of 68% of variation explained. Though this model captures evident demand behaviour the forecasting errors still exist. For a baseline model it performed well.

The XGBoost improved the performance slightly compared to the Random Forest model and the results are

- MAE = 1.3010
- RMSE = 2.3910
- $R^2 = 0.6860$

On comparison the error values are low and slightly higher explanatory power. This indicates our previous point that sequential learning may better at capturing retail demand patterns as it learns from previous prediction errors (Chen and Guestrin, 2016, pp. 785–794).

After the first XGBoost model hyperparameter tuning and business features were added to check whether it improves the efficiency of the model performance the result were as follows :

- MAE = 1.3010
- RMSE = 2.3899
- $R^2 = 0.6863$

The improvement in performance was marginal . Although the gains are small the business variables still manage to contribute additional predictive value. The possible reason for this outcome might be that the retail demand had been already strongly explained by the historical sales behaviour.

Overall, the optimized XGBoost is identified as the best performing model though the difference is relatively small stating that the business feature affected the forecasting performance in a very moderate way.

Model comparison table :

Model	MAE	RMSE	R ²
Random Forest	1.3079	2.4034	0.6827
XGBoost	1.3010	2.3910	0.6860
Optimized XGBoost	1.3010	2.3899	0.6863

5.2 Interpretation of Forecasting Results

The forecasting results are interpreted to understand why the models performed this way , what are all the factors influenced the predictive performance of the models , whether or not the features that added contributed to the model meaningfully. And the results indicate that the machine learning model used in the research were able to capture the retail demand behaviour reasonably well. The R square value (68 %) supports that statement. However, there are still prediction errors which can be caused because the retail demand is volatile , uncertain and complex in nature (Chopra and Meindl, 2019, pp. 184–190).

The main reason behind the strong forecasting performance is likely the historical demand features especially

- lag_7
- rolling_mean_7

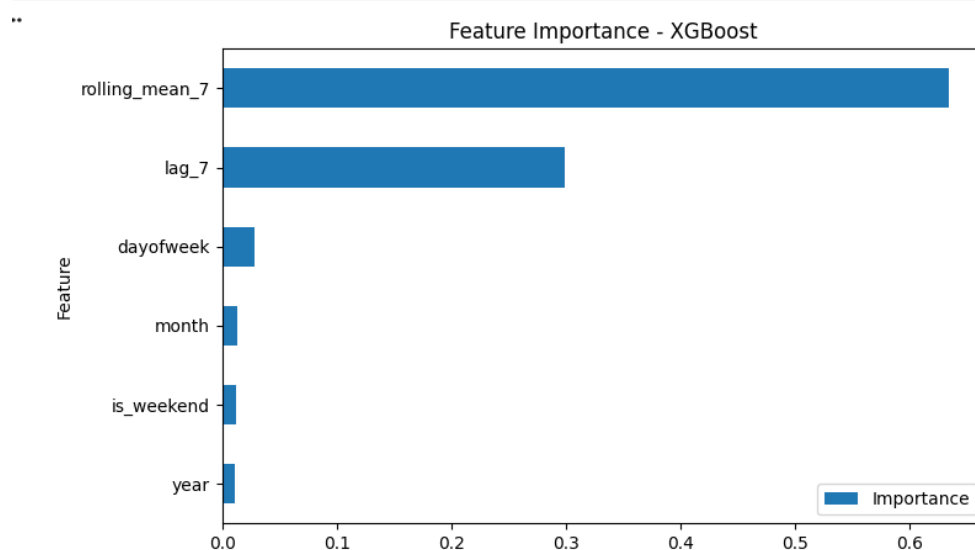
This is supported by the EDA of this research where it is revealed that there is weekly recurring demand behaviour and there are daily fluctuations. As mentioned earlier past sales strongly influence future demand. One possible reason behind the better performance of XGBoost model is that the purchasing pattern of food products often repeats weekly. All these findings suggest that the historical demand information plays a dominant role in short-term retail forecasting.

As mentioned above the business features such as SNAP CA and event encoded improved forecasting only to a marginal level. This is due to historical sales behaviour heavily impact on demand already and the influence of business features may occur only during specific time periods rather than continuously. Therefore, only limited overall improvement is found (Chopra and Meindl, 2019, pp. 184–190).

In this research 1 day-ahead forecasting horizon was used as the short-term forecasting is high useful for inventory planning , stock replenishment and food demand forecasting. It also reduces the stockout risk and overstocking especially food product which is considered as a crucial category in retail stores. However very short horizon may naturally produce smaller differences in between models (Chopra and Meindl, 2019, pp. 184–190).

The results indicate XGBoost is effective for short-term retail demand forecasting especially when historical sales behaviour is incorporated. The forecasting performance was reasonable but not perfect because the retail demand remains still uncertain and influenced by multiple external factors.

5.3 Feature Importance analysis



Feature importance analysis is conducted to find out which variable is influencing the most in the performance of the model, to understand what drives prediction. It also helps to evaluate the usefulness of the engineered features by providing feature important scores.

The top feature that we can see in the graph is `rolling_mean_7` = 0.63 which suggests the recent average sales strongly influence the model predictive performance of future demand. As discussed in the EDA segment that weekly recurring demand

pattern were found which means that the short-term demand behaviour is important in food retail forecasting as the purchasing pattern is often repetitive.

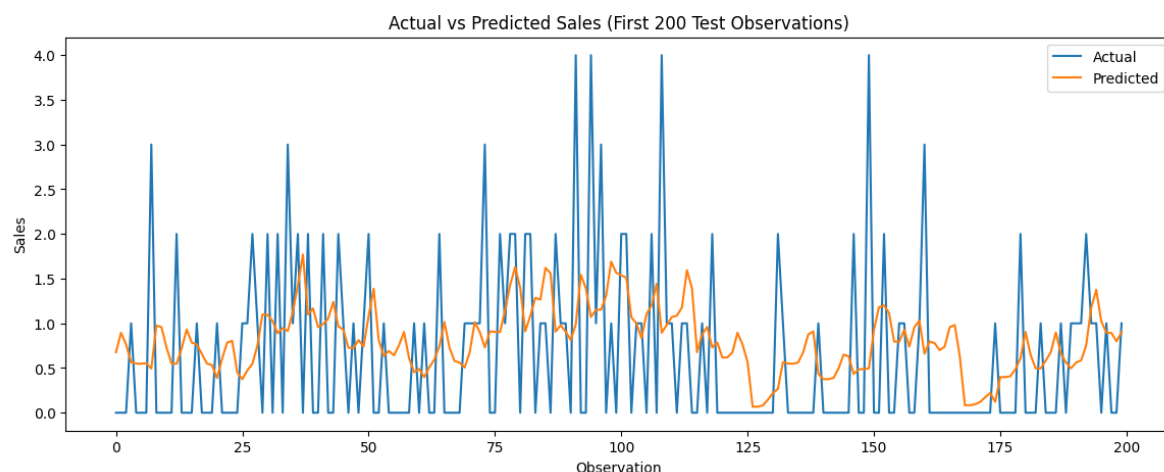
Followed by `rolling_mean_7` we have `lag_7` as the second most important feature to influence the model in its predictive performance (0.30). In simple terms sales from seven days prior have helped the model to forecast in a reasonable way. This is also due to the weekly purchasing behaviour being repeated again and again as shown in the EDA findings.

Then the final smaller feature such as day of the week, month, weekend, year have a marginal impact on the model in its forecasting process. Although these variables contributed less to the prediction power of the model, they still provided additional temporal context for forecasting. This might be due to previous finding that the historical data variables affect the demand in a major way. Therefore, these calendar variables play a supporting role in the forecasting process.

Business related features were added in optimizing the XGBoost model to see whether it provides additional predictive contribution; however, the improvement of the model was marginal.

Overall, the feature importance analysis suggests that the historical demand variables play a major role in influencing the forecasting process of the model, the temporal and business-related variables stood as a supportive role to the model analysis.

5.4 Actual vs Predicted Analysis

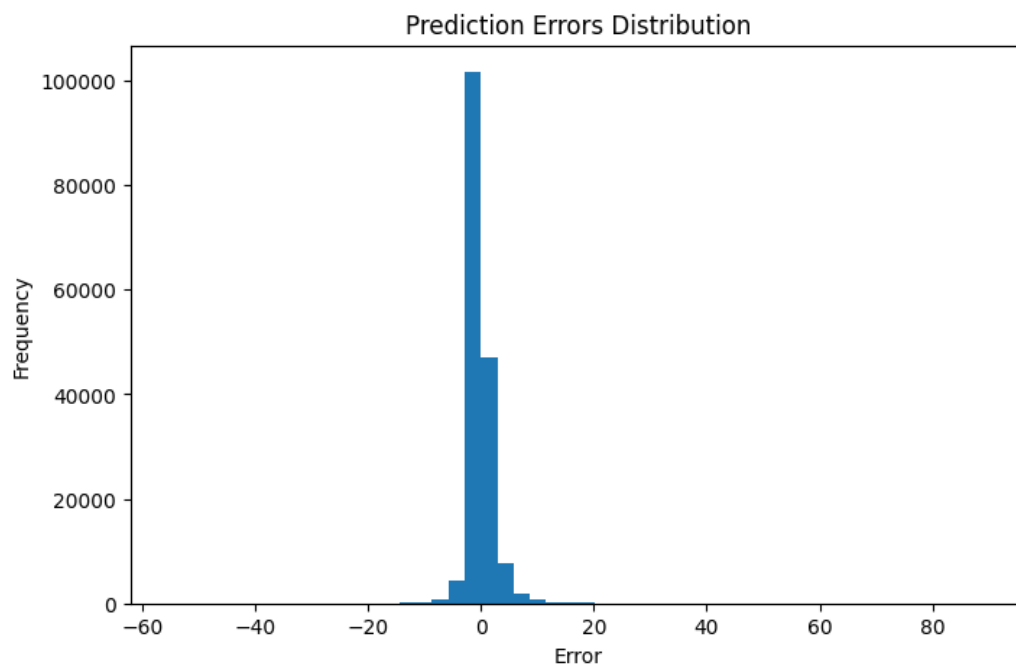


This visual graph was conducted to compare the actual sales with the predicted sales which we had found with our model. It also helps to assess the forecasting behaviour visually. The purpose is to determine whether the model used for the forecasting of demand actually captures the overall sales movement and demand fluctuations.

As shown in the graph the predicted values tend to follow the overall movement of the actual sales capturing general trend at a reasonable rate. The noticeable point is that the predicted values are actually smoother than the actual sales values by this it means the forecasting model captures the overall demand direction but not every sudden fluctuation.

Some peaks and sudden changes are not perfectly captured , the possible reason behind it is that the retail demand is naturally affected by randomness , unexpected events and temporary changes in the behavioural patterns of the customers. Despite these deviations the model still manages to provide reasonable forecasting capability especially for short-term demand forecasting. This level of forecasting can still be useful for inventory management and shot-term decision making in the retail sectors.

5.5 Residual Analysis



The residual analysis is conducted to analyse the forecasting errors , to understand the prediction behaviour and asses the reliability of the model which is used for forecasting the demand. The term residual here means the difference between actual sales and the predicted sales. There are two main purposes for this analysis first, to check whether the forecasting follows a reasonable pattern and second, whether the prediction stay close to the actual values(James et al., 2021, pp. 254–260).

The residual distribution in the graph is roughly concentrated around zero near centre of distribution which suggests that the predicted values generally remain close to

the actual sales and majority of prediction errors relatively small. This also suggest that the model performs reasonably well for demand forecasting.

It is also found that the residual spread exists on both sides of the centre stating that model sometimes overpredicts and sometime underpredicts the demand. The possible reason behind this can be sudden demand fluctuations , temporary behaviour changes and unexpected influences in the retail sector.

Although there is uncertainty in the prediction process of the model , it is still reasonably reliable for short-term retail forecasting. This analysis supports the reliability and practical usefulness of the forecasting model while also showing that the retail demand is naturally uncertain and fluctuating.

6. Business Insights and Recommendations

6.1 Business Implications

Retail demand forecasting is important because it helps business to plan the inventory , reduce uncertainty in business and helps improve decision making in the retail sector (Chopra and Meindl, 2019, pp. 184–190). Poor forecasting may lead to stockouts - which affects the sales , overstocking - which increases the management cost and wastage due to expiry of unsold goods. This is specifically important for the food category as it is the frequently purchased product , perishable and sensitive to demand changes.

The finding of the machine learning models in this research shows reasonable effect on short-term retail demand forecasting. The optimized XGBoost is identified as the best performing model in the research. The reasons are lowest RMSE , lowestMAEand the highest R Square compared to other models which are conducted in this research. However , the improvement is moderate rather than a dramatic improvement.

The practical application of this model if it is used by the retailers are to plan the inventory daily, stock replenishment , reducing food shortages by forecasting the demand. Especially 1-day-ahead forecasting helps short-term operational decisions.

The feature important analysis showed that the historical sales behaviour is the most important variable to influence the predictive capacity of the model especially , `rolling_mean_7` and `lag_7`. This means that the retail businesses should focus on monitoring the historical sales carefully as the past demand strongly influence the future sales.

The business features such as SNAP programme and event variables provided the additional value and suggest external factors also should be considered though it

may only impact specific periods . In business terms the retailers can benefit from tacking holidays , special events such as super bowl , promotions and government policy periods as the demand surges during those times.

The forecasting cannot eliminate the uncertainty in demand completely. It is not perfect because the retail demand is influenced by consumer behaviour which can change , unexpected events and change in market conditions.

This research demonstrate that the machine learning forecasting can support retail decision making and improve the efficiency of the inventory management particularly for the short-term food demand forecasting.

7. Conclusions

7.1 conclusion

The purpose of this research was to develop a retail demand forecasting model using machine learning techniques on the M5 competition forecasting dataset which was restricted to a particular store and category (CA_1 , FOODS) to make the research manageable and focused.

The research had the workflow starting from preprocessing the data , conducting EDA , feature engineering , development of models , hyperparameter tuning and evaluation of the models which includes comparison with different model which were created.

The EDA revealed that there is weekly recurring demand (a repetitive pattern), moderate seasonal demand pattern , high increase in sales during the weekends , moderate influence of SNAP and event-driven demand changes. These finding influenced the feature engineering to include historical demand , temporal, SNAP and even encoded variables.

Three models were created Random Forest , XGBoost and optimized XGBoost , where optimized XGBoost is identified as the best performing model though the improvement is moderate.

The feature importance analysis revealed that the rolling_mean_7 and lag_7 influenced the most in the performance of the model suggesting that the historical demand behaviour dominated the short-term forecasting in the retail sector.

The research demonstrates that machine learning forecasting is useful for the inventory planning , stock replenishment and supporting retail decision making especially in forecasting the demand for the short-term in food category.

Although the forecasting uncertainty cannot be eliminated completely , the research demonstrates that the machine learning models can support in predicting the demand for the short – term and decision making practically.

7.2 Limitations of the research

The first limitation is that the data is restricted to one particular store(CA_1) and one category (FOODS) to reduce the complexity in data for computation but the results may not generalise to other stores ,other product categories and if it is a wide retail environment.

The research used 1-day-ahead forecasting horizon , the limitation of using this it focuses primarily on predicting short-term demand and may not be applicable for the long-term forecasting challenges such as weekly demand planning and monthly inventory decisions but still will be useful for daily retail operations.

Hyperparameter tuning conducted through manual experimentation and without the implementation of the grid search or cross-validation. The reason behind this is time and computational limitations.

The categorical variables are encoded using label encoding which may create an artificial order between categories, but it is less sensitive for the tree-based models as mentioned above. The alternative for that method can be Catboost or one-hot encoding which could improve future research.

Despite all these limitations the research still provides valuable insights into the short-term retail demand forecasting.

7.3 recommendation for future research

The future research can be done in an expanded dataset beyond a certain store or category including multiple stores , multiple states and multiple product categories. This will improve the generalisation of the findings and will be applicable on a long retail environment.

The future studies may test longer forecasting horizon such as 7-day , 14-day or monthly forecasting to understand the long-term forecasting capability which will be useful for strategic inventory planning , decision making on the supply chain for a longer period of time.

The future research may implement grid search , random search , cross-validations for the hyperparameter tuning to identify better performing parameter combinations and could potentially improve the forecasting accuracy.

Further research may compare Catboost , LSTM , Prophet, ARIMA with the created XGBoost model to capture the retail demand differently. It may include additional business variables such as price changes , promotions , economic indicators, weather, competitor activity to understand the external demand drivers more efficiently.

Finally, the future research can improve the forecasting accuracy and it will be more useful for the retail sector as the research will be done by expanding scope , models , forecasting period and external variables.

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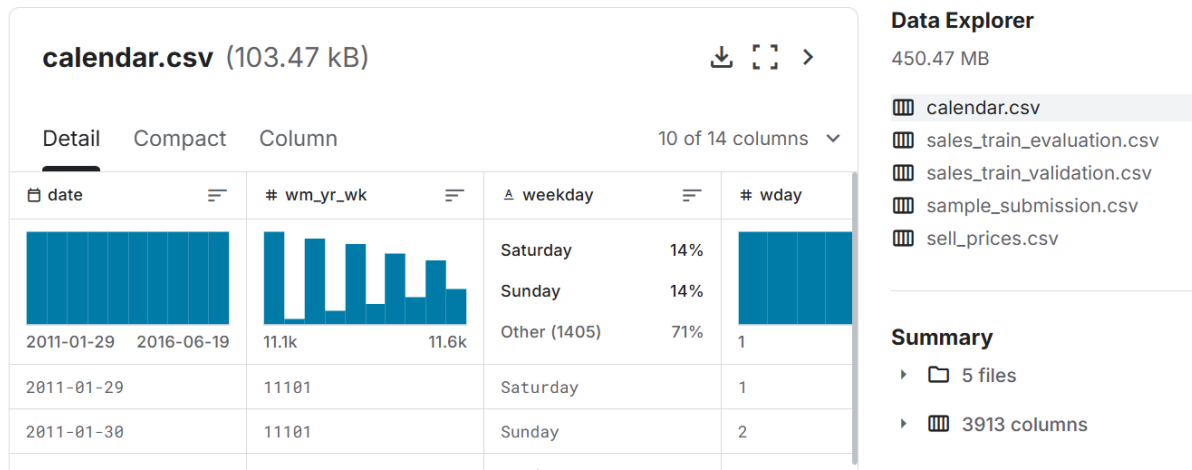
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Appendix

Appendix A: M5 Forecasting Dataset and Data filtering

<https://www.kaggle.com/competitions/m5-forecasting-accuracy>



```
sales_ca1 = sales[sales['store_id'] == 'CA_1']
```

```
🎬 sales_ca1_foods = sales_ca1[sales_ca1['cat_id'] == 'FOODS']
```

```
sales_ca1_foods.shape
```

```
(1437, 1919)
```

```
sales_ca1_foods.to_csv("filtered_sales.csv", index=False)
```


Appendix B: Data Transformation and Preprocessing

Wide → Long format transformation

```
) id_vars = ['id', 'item_id', 'dept_id', 'cat_id', 'store_id', 'state_id']

long_df = pd.melt(
    sales_cal_foods,
    id_vars=id_vars,
    var_name='d',
    value_name='sales'
)
```

Merge with calendar dataset

```
) id_vars = ['id', 'item_id', 'dept_id', 'cat_id', 'store_id', 'state_id']

long_df = pd.melt(
    sales_cal_foods,
    id_vars=id_vars,
    var_name='d',
    value_name='sales'
)
```

Date conversion

```
long_df['date'] = pd.to_datetime(long_df['date'])
```

```
long_df['dayofweek'] = long_df['date'].dt.dayofweek
long_df['month'] = long_df['date'].dt.month
long_df['year'] = long_df['date'].dt.year
long_df['weekofyear'] = long_df['date'].dt.isocalendar().week.astype(int)
long_df['is_weekend'] = long_df['dayofweek'].isin([5,6]).astype(int)
```

Temporal feature creation

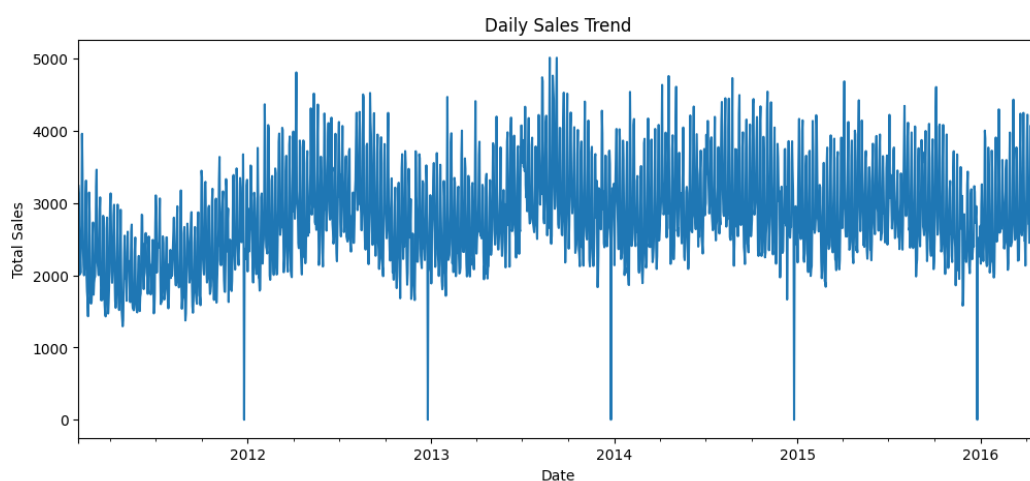
```
long_df[['date', 'dayofweek', 'month', 'year', 'is_weekend']].head()
```

	date	dayofweek	month	year	is_weekend
0	2011-01-29	5	1	2011	1
1	2011-01-29	5	1	2011	1
2	2011-01-29	5	1	2011	1
3	2011-01-29	5	1	2011	1
4	2011-01-29	5	1	2011	1

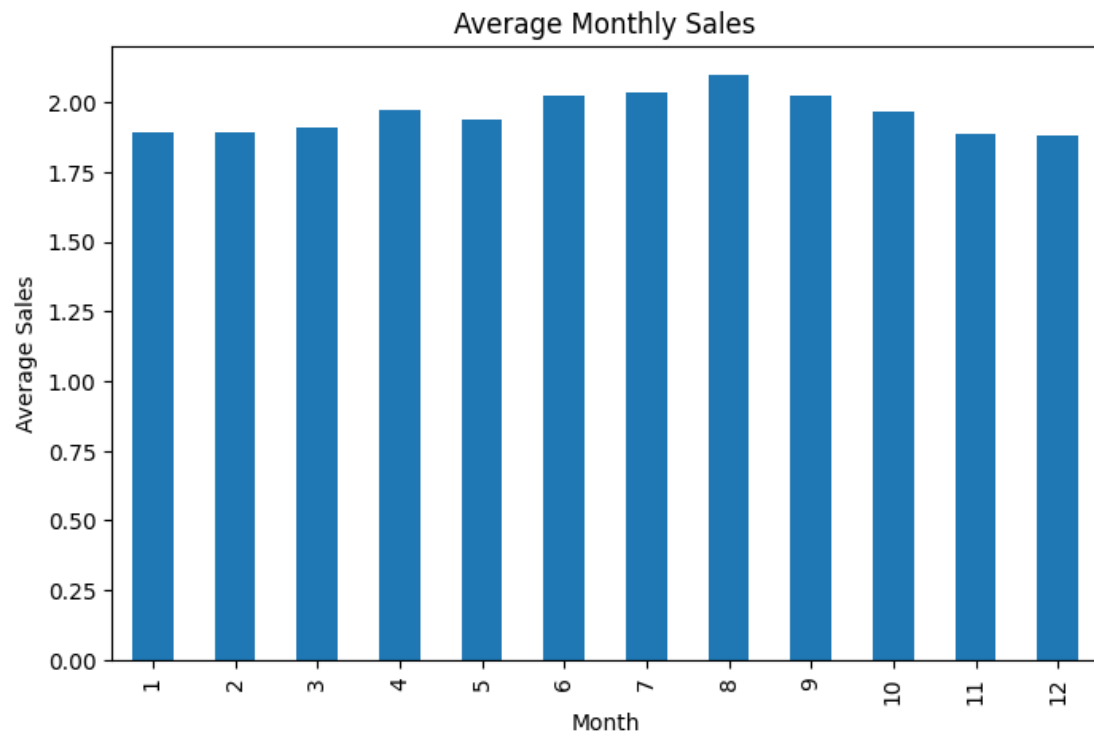
```
long_df['date'].nunique()
```

1913

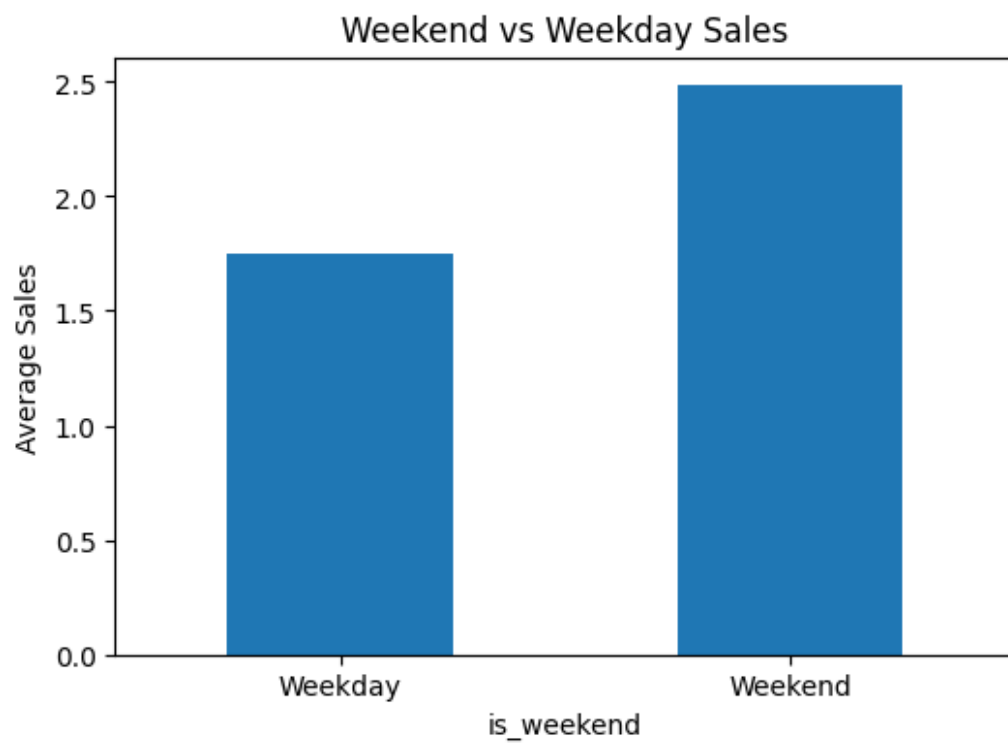
Appendix C: Exploratory Data Analysis (EDA)



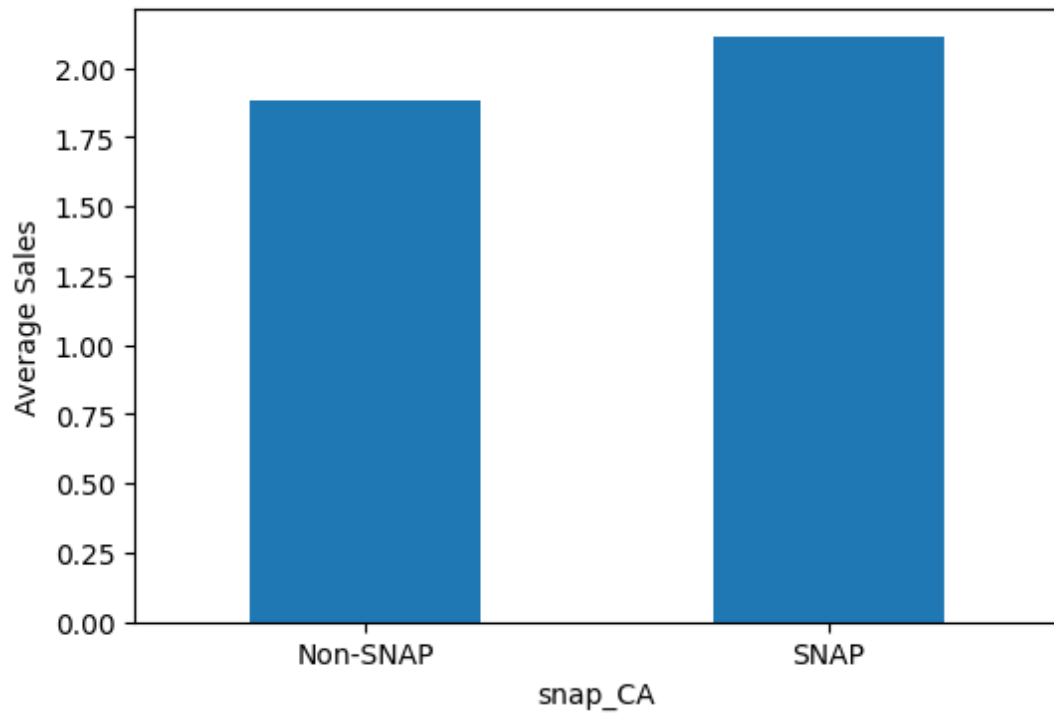
.



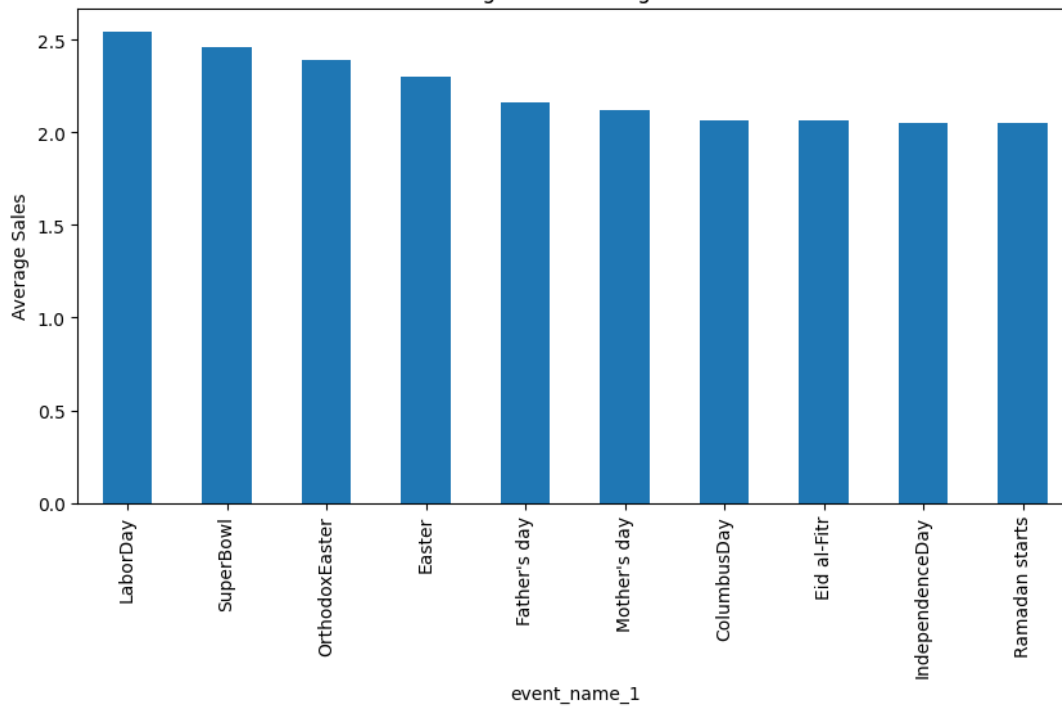
..



SNAP vs Non-SNAP Sales



Average Sales During Events



Appendix D: Feature Engineering

Sorting for time-series

```
long_df['date'].sort_values().unique()[:10]
```

```
<DatetimeArray>  
['2011-01-29 00:00:00', '2011-01-30 00:00:00', '2011-01-31 00:00:00',  
 '2011-02-01 00:00:00', '2011-02-02 00:00:00', '2011-02-03 00:00:00',  
 '2011-02-04 00:00:00', '2011-02-05 00:00:00', '2011-02-06 00:00:00',  
 '2011-02-07 00:00:00']  
Length: 10, dtype: datetime64[ns]
```

Lag feature

```
long_df = long_df.sort_values(['id', 'date'])  
  
long_df['lag_7'] = long_df.groupby('id')['sales'].shift(7)
```

Rolling mean feature

```
long_df['rolling_mean_7'] = (  
    long_df.groupby('id')['sales']  
    .transform(lambda x: x.shift(1).rolling(7).mean())  
)
```

Appendix E: Model Development

Feature selection

```
features = [  
    'dayofweek',  
    'month',  
    'year',  
    'is_weekend',  
    'lag_7',  
    'rolling_mean_7'  
)  
  
X = model_df[features]  
y = model_df['sales']
```

Train-test split

```
split_date = '2016-01-01'

X_train = X[model_df['date'] < split_date]
X_test  = X[model_df['date'] >= split_date]

y_train = y[model_df['date'] < split_date]
y_test  = y[model_df['date'] >= split_date]
```

Random Forest Forecasting Model

```
) rf = RandomForestRegressor(
    n_estimators=50,
    max_depth=10,
    random_state=42,
    n_jobs=-1
)

rf.fit(X_train, y_train)
```

XGBoost Forecasting Model

```
xgb = XGBRegressor(
    n_estimators=100,
    max_depth=6,
    learning_rate=0.1,
    subsample=0.8,
    colsample_bytree=0.8,
    random_state=42,
    n_jobs=-1
)

xgb.fit(X_train, y_train)
```

Optimized XGBoost Model

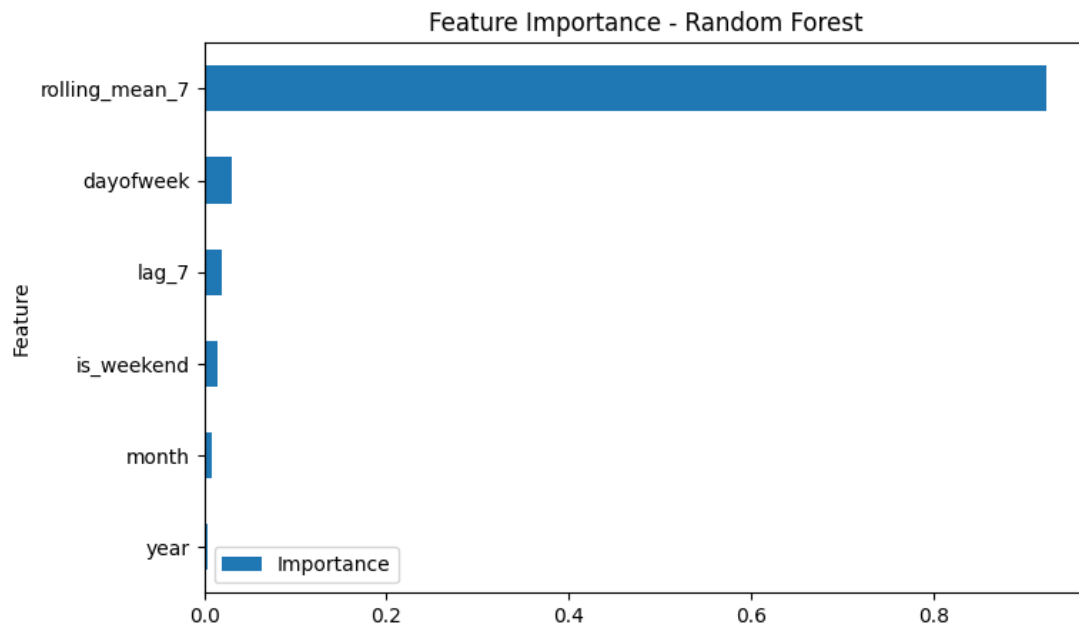
```
xgb_optimized = XGBRegressor(  
    n_estimators=200,  
    max_depth=8,  
    learning_rate=0.05,  
    subsample=0.8,  
    colsample_bytree=0.8,  
    min_child_weight=3,  
    random_state=42,  
    n_jobs=-1  
)  
  
xgb_optimized.fit(X_train, y_train)
```

Appendix F: Model Evaluation and Results

Random Forest Model Evaluation

```
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score  
import numpy as np  
  
mae = mean_absolute_error(y_test, preds)  
rmse = np.sqrt(mean_squared_error(y_test, preds))  
r2 = r2_score(y_test, preds)  
  
print("MAE:", mae)  
print("RMSE:", rmse)  
print("R2:", r2)
```

```
MAE: 1.3078982038257156  
RMSE: 2.403441593686039  
R2: 0.682709282192339
```

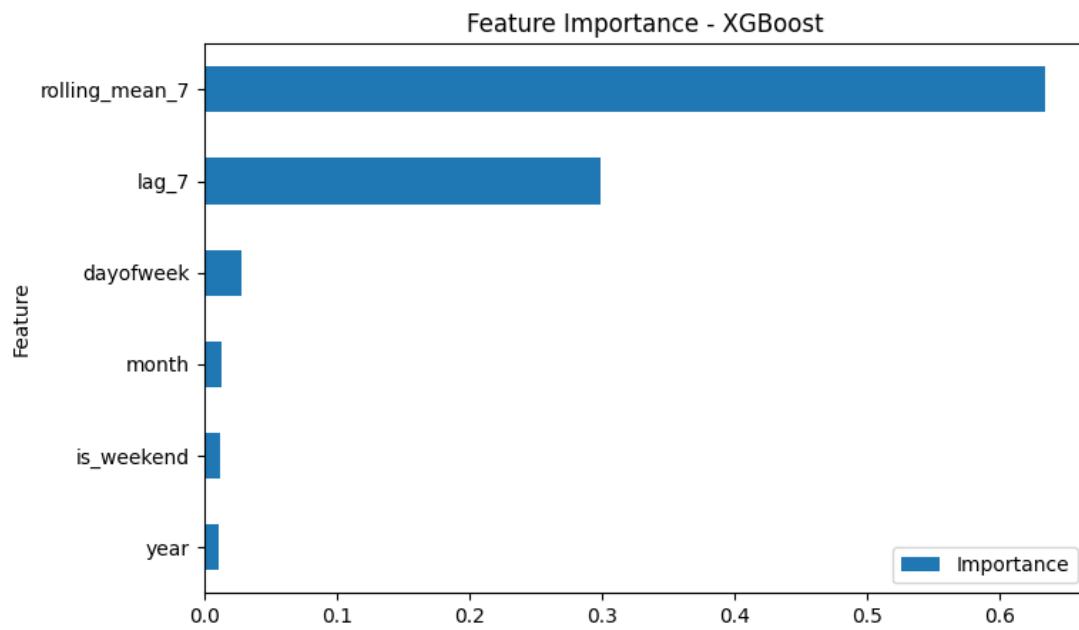


XGBoost Forecasting Model evaluation

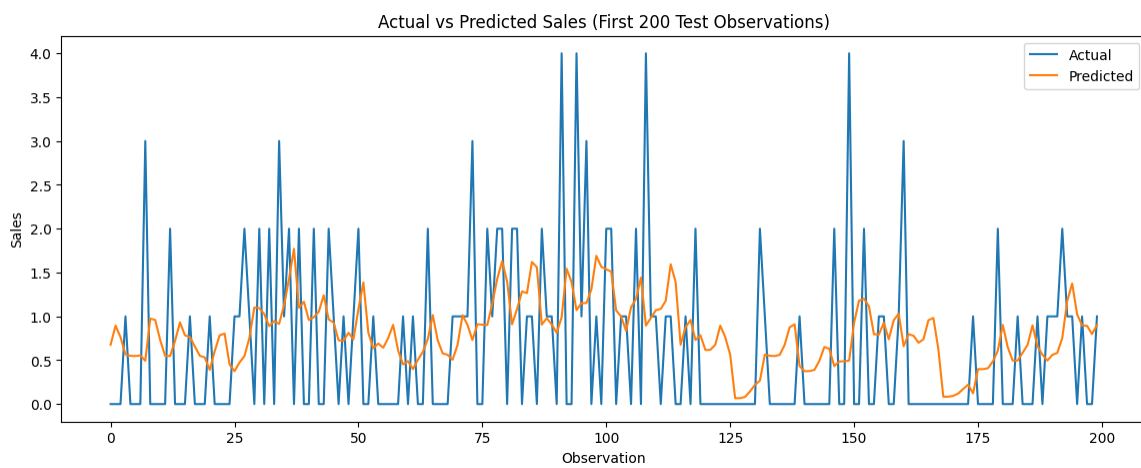
```
| mae = mean_absolute_error(y_test, xgb_preds)
  rmse = np.sqrt(mean_squared_error(y_test, xgb_preds))
  r2 = r2_score(y_test, xgb_preds)

  print("XGBoost MAE:", mae)
  print("XGBoost RMSE:", rmse)
  print("XGBoost R2:", r2)
```

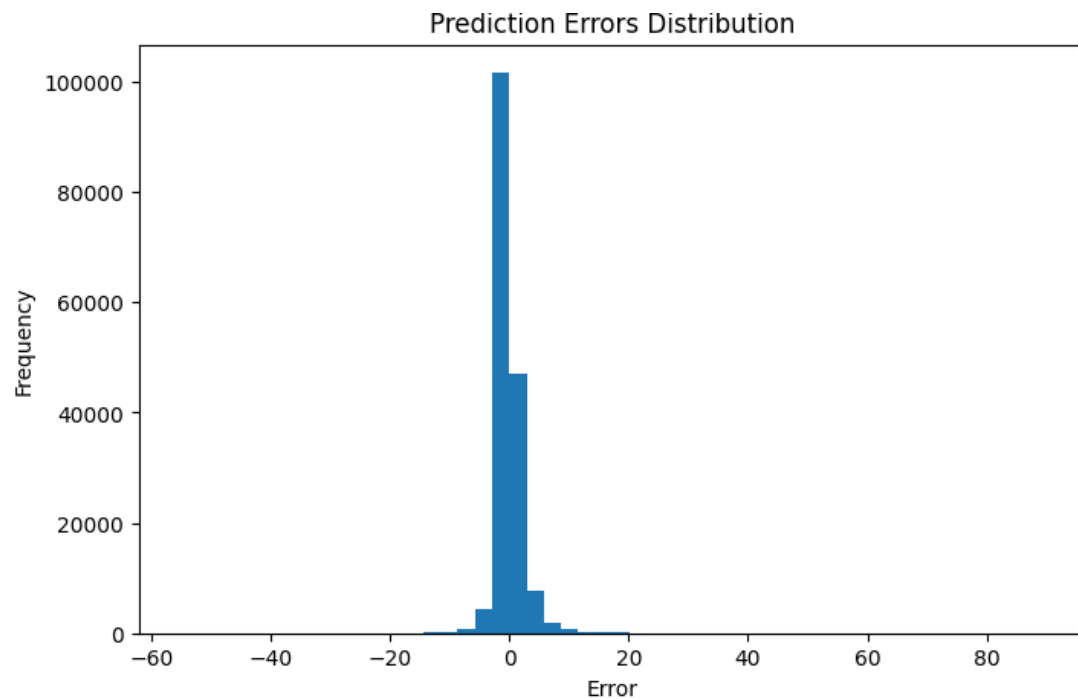
```
XGBoost MAE: 1.3010481595993042
XGBoost RMSE: 2.3910402423530552
XGBoost R2: 0.685975193977356
```

Actual VS predicted sales



Prediction error distribution



Appendix G: GitHub Repository

The complete implementation code, preprocessing workflow, feature engineering process, exploratory data analysis and machine learning model development used in this research are available through the GitHub repository below:

GitHub Repository Link: <https://github.com/KarthikRamu-gif/retail-demand-forecasting-ml>

Appendix h: AI use declaration

During the course of this research study, Artificial Intelligence (AI) tools were used for purposes that were extremely limited in scope. In addition, ai was used as an assistive tool to help with grammatical revisions, enhance the quality of sentences; enhance clarity of language; guide the organization of documents and formatting of reports.

All research design decisions, data preprocessing, exploratory data analysis (EDA), feature engineering, machine learning model development, model evaluation, interpretation of findings and coding implementation were independently conducted by the researcher.

The Python implementation, machine learning model construction, hyperparameter tuning and forecasting analysis were personally developed and validated by the researcher using the M5 Forecasting Dataset.

